VIET NAM NATIONAL UNIVERSITY HO CHI MINH CITY

UNIVERSITY OF ECONOMICS AND LAW

FACULTY OF FINANCE AND BANKING

GRADUATION THESIS

BUILDING A PAYMENT RECONCILIATION SYSTEM FOR AFFILIATE MARKETING ACTIVITIES: A CASE STUDY ON THE TIKTOK SHOP E-COMMERCE PLATFORM

Supervisor : **Assoc. Prof. Nguyễn Anh Phong**

Student name : **Bùi Võ Kim Ngân**

Student ID : **K214140945**

Class : **K21414A**

Ho Chi Minh City, March 2025

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INSTRUCTOR'S COMMENT

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LIST OF ABBREVIATIONS

Abbreviation Symbols	**Full Text**
AMNSTE	Affiliate Marketing Network simulation and testing environment
API	Application Programming Interface
BPMN	Business Process Modeling Notation
CPA	Cost per action
CPC	Cost per click
CPL	Cost per lead
CPS	Cost per sale
DFD	Data flow diagram
ESS	Electronic Security System
FAM	Functional Architecture Model
PPS	Pay per sale
RSA	Rivest–Shamir–Adleman
SME	Small and Medium enterprises
SSL	Secure socket layers
TAM	Technology Acceptance Modal
TAP	Tiktok Affiliate Partner
TDES/3DES	Triple Data Encryption Standard
TTP	Trusted Third Party
UML	Unified Modeling Language

ABSTRACT

The research aims to examine the traditional commission payment reconciliation procedure by conducting interviews with industry experts and analyzing data from more than 5,900 orders between September 1, 2024, and January 31, 2025, in affiliate marketing operations. With the boom of e-commerce platforms like Shopee and TikTok Shop, affiliate marketing is becoming increasingly well-known, attracting many participants. Affiliate marketing is a mechanism that has facilitated the success and widespread recognition of products on e-commerce platforms. However, managing this activity in the affiliate marketing network still faces many challenges, such as security, stability, and privacy (Marten Weijer, 2019). Participants encounter difficulties in tracking, monitoring, and identifying fraudulent behaviors. Based on these practical needs, the author analyzes the characteristics of the traffic, the number of orders generated from the program, and the proportions of marketing methods publishers use in the network. As a result, a suitable reconciliation system is proposed based on a Use Case Diagram, Data Flow Diagram, and Business Process Modeling Notation (BPMN).

Keywords: E-commerce, Affiliate marketing, Commission, Reconciliation System.

1. INTRODUCTION

1.1. Statement of the problem

In recent years, e-commerce has emerged as one of the most promising markets, and its value is steadily increasing. The estimated global e-commerce market value for 2024 is approximately $5 trillion, with a growth rate of 15% per year from 2019 to 2024 (ECDB, 2024). E-commerce is the sale, and transfer of goods, services, and information over the Internet, allowing access and ordering from various locations without the need for physical presence at the point of sale. Participants in the e-commerce platform may include business partners, customer support, and other non-commerce-related jobs. Apriani et al. (2024) claimed that utilizing e-commerce in the SMEs' operation is crucial. E-commerce platforms have enhanced the financial capabilities of SMEs by boosting product visibility and engagement levels, which has a beneficial effect on the financial outcomes of the businesses involved (Guo et al., 2024).

In Vietnam, sales on e-commerce platforms are becoming increasingly popular. Hoan et al. (2024) discovered Vietnam's e-commerce market based on data from Metric.vn was the second largest in Southeast Asia in the first quarter of 2022, only behind Indonesia. In 2024, the e-commerce sector experienced growth exceeding 25%, reaching a scale of over $25 billion (Vietnam E-commerce Association, 2024). One of the methods that have helped products on e-commerce platforms gain popularity and widespread recognition is affiliate marketing. Based on the shopping behaviors of Vietnamese consumers, revenue from the affiliate marketing sector in 2022 accounted for nearly 77% of the total e-commerce revenue, with TikTok emerging as the leading platform, representing over 42% of the market share (Accesstrade, 2022).

The growing volume and value of transactions necessitate the automation of reconciliation and payment procedures in affiliate marketing to satisfy the swift, ongoing, and transparent requirements of management entities. Nevertheless, research on commission reconciliation between publishers and affiliate marketing networks in Vietnam is limited. Research in Vietnam has primarily focused on integrating affiliate marketing activities between e-commerce platforms and networks (Nguyễn Thị Kim Tuyến & Đàm Thị Phương

Thảo, 2019; Phạm Xuân Lan, 2023) or exploring the development potential of the domestic market (Phạm Thị Huyền Kiều et al., 2021). However, there is a lack of studies addressing the establishment of reconciliation systems between publishers and networks. Therefore, this research will study an automated commission reconciliation and payment system, utilizing data on products, content creators, and order payment information recorded through the API protocol of TikTok Affiliate Partner. The system will automatically calculate affiliate commission fees based on each order paid through various methods. Additionally, the research will evaluate the performance of publishers throughout each campaign period, facilitating the recommendation of appropriate management and development plans for MCNs and enterprises.

1.2. Research objectives

1.2.1. General objective

The research evaluated the traditional reconciliation and payment processes used by management entities overseeing affiliate marketing campaigns. Based on this evaluation, the study proposes an automated reconciliation and commission payment system for affiliate marketing activities. This approach aims to optimize management time, oversight, and overall operational efficiency for businesses. Additionally, the research analyzes the performance of content creators and suggests appropriate development strategies for the management system.

1.2.2. Specific objectives

Analysis of the Current Payment and Reconciliation Process: Analyzing and evaluating traditional methods for content creators. Then, identifies existing issues and limitations.

Performance Analysis of Content Creators in Affiliate Marketing: Examining order placement, return rates, successful payments, and consumer characteristics.

Examination of Affiliate Marketing Methods and Commission Payments: This section investigates affiliate marketing strategies and commission payments, monitoring cash flow, revenue, costs, and commissions paid to content creators and management.

Proposed Solutions for Commission Reconciliation System: Proposes solutions for developing a commission reconciliation system, tailored to order status, payment methods, and transaction conditions.

1.3. Research questions

How does the traditional payment and reconciliation process operate for different payment methods?

What characteristics do orders from content creators participating in the payment and reconciliation system typically exhibit?

How does the system's automation solution work? Do these solutions meet the purchasing characteristics of affiliate marketing activities on the TikTok Shop e-commerce platform?

How can the effectiveness of these solutions be assessed, and what recommendations can be made to enhance the efficiency of the process?

1.4. Subject and Scope of study

Subject of study

This study is conducted based on the needs of affiliate marketing networks, such as agencies and Multi-Channel Networks (MCNs), to manage commissions, the reconciliation process, and payments for content creators or influencers active on the TikTok Shop e-commerce platform when marketing products.

Scope of study

The data sample provided by the organization spans five months, encompassing over 5,900 orders from September 1, 2024, to January 31, 2024. The data includes order information, product details, and publicly available information about content creators on the TikTok platform. These orders represent actual transactions and can be regarded as a representative sample for evaluating the process and the effectiveness of the established system. The author has also assessed the timeliness and transparency of the provided data about the information publicly posted on the TikTok platform.

1.5. Research Methods

Qualitative Method: The qualitative method involves a comprehensive literature review, surveying needs, and analyzing the management process of the reconciliation and payment system in the organization. This approach aims to address the objectives outlined in the research goals.

Quantitative Method: The quantitative method evaluates the characteristics of orders and the performance of content creators. Additionally, the study examines statistics related to the number of transactions, revenue, and marketing costs the organization will incur. The research also analyzes the marketing methods, payment processes, and order fulfillment times. This analysis will inform the proposal of an appropriate reconciliation and payment system.

2. LITERATURE REVIEW

2.1. Theoretical Literature Review

2.1.1. Affiliate Marketing

Affiliate marketing is a form in which a referrer receives a corresponding commission upon successfully promoting a product to another person. This method is also referred to as "Performance Marketing" where partners strive to attract customers to visit the merchant's website through referral links. When customers purchase products on the platform, the partners receive a corresponding commission (Ivkovic and Milanov, 2010). Affiliate marketing is an effective marketing strategy for increasing sales without incurring substantial initial costs. It is considered one of the most influential forms of marketing across various sectors, including entertainment, e-commerce, banking, and finance.

The implementation of affiliate marketing programs on e-commerce platforms greatly depends on customer trust. Rousseau et al. (1998) claimed that trust is a psychological state formed by accepting risk based on expectations about the behavior or motives of others. Trust leads a person to expect that others will act in their favor, despite the inability to control the actions of others (Moorman et al., 1992; Morgan and Hunt, 1994). When customers make purchases through affiliate marketing programs, it signifies that they trust their referrers. Impulse buying behavior among TikTok users indicates that influencers, personal branding, and affiliate marketing activities significantly impact users' purchasing decisions (Andriani & Associates, 2024). Affiliate marketing can also become a primary form of marketing on e-commerce platforms, and achieving success in this area requires optimizing the relationship between the referrer and the affiliate partner (Duffy, 2005).

Currently, the payment structures for affiliate marketing programs are quite diverse. According to Hossan and Ahammad (2013), the compensation for commissions in these programs can be classified based on cost per sale (CPS) or pay per sale (PPS). This is an affiliate model where the commission percentage is determined by the number of orders sold through advertising and marketing. Additionally, it may also be based on specific customer actions, such as software downloads or newsletter sign-ups (Cost per Action - CPA), cost per lead (CPL), or cost per click (CPC) is a concept that enables referrers to get compensation depending on the frequency of customer clicks on the referral link. Depending on the stipulated conditions, referrers will be compensated according to the model they utilize for marketing and affiliation.

Moreover, Libai et al. (2003) classified affiliate marketing programs into two distinct types: one-to-one and one-to-many programs. In one-to-one programs, publishers enter into exclusive contracts that define the terms and conditions between the parties. In contrast, in one-to-many relationships, publishers disseminate referral links from e-commerce platforms such as Amazon. Upon consumer clicks and subsequent purchases, the referrer earns a percentage of the revenue. Currently, one-to-many programs are applied on most e-commerce platforms such as Shopee and TikTok Shop, providing substantial income for participants.

The affiliate marketing system consists of entities such as Merchants who sell products through online advertising programs, Publishers who promote products to consumers through affiliate marketing, and Networks which are third parties connecting merchants and publishers. Within this system, networks are responsible for tracking, managing, and reporting affiliate marketing programs through clicks or purchases made via referral links. Furthermore, networks provide effective solutions for monthly commission payments in affiliate marketing programs to publishers (Edelman & Brandi, 2015).

Affiliate marketing's applicability extends beyond e-commerce platforms, developing in other areas such as banking and fintech. Research by Martynov (2023) on marketing activities as part of customer acquisition strategies in banks found that most banks in Russia implement these marketing programs, primarily receiving commissions based on the CPA model. The positive signals from the CPA model also provide numerous opportunities for improvement in experimental models such as CPL, benefitting the revenues of banks and other financial institutions.

2.1.2. Reconciliation

Reconciliation in the financial system is the process of matching and verifying transactions to ensure the accuracy, integrity, and validity of payments. VAN Schaik & Frans DJ (2023) stated that the primary objective of reconciliation is to explain the discrepancies between budgeted revenues and expenses compared to recognized revenues and expenses, or cash inflows and outflows. The reconciliation procedure is essential for ensuring consistency, improving comprehension, and augmenting the trustworthiness of financial reporting.

In a research on the reconciliation of check payments, Stickney (1966) argues that the processes of “payment” and “reconciliation” of checks involve controlling and verifying the authenticity and validity of the checks. Simplifying processes can produce substantial advantages with the development and execution of control systems, while also improving the potential for a cashless economy. However, it may also pose challenges for organizations that adhere to traditional processes.

Schultze (2018) identified three categories of transactions in the reconciliation process within commerce that can be recorded and processed electronically, on paper, or in a hybrid format: invoices, purchase orders, and shipping documents. Following the reconciliation process, the output will consist of balanced and accurate data (Turner & Associates, 2017). Traditional reconciliation necessitates substantial accounting resources, and frequently, the reconciliation for MSMEs is conducted separately, reprocessing data from e-commerce platforms and amalgamating it with current transactions before the preparation of financial reports (Deloitte, 2018). This process typically demands numerous resources and manual reconciliation, which can lead to human errors in recording and data consolidation, resulting in duplication issues and affecting the operational efficiency of businesses (Inampudi & Associates, 2023). To mitigate errors from manual reconciliation, organizations should consider adopting automated reconciliation methods utilizing technology applications.

2.2. Empirical Literature Review

2.2.1. Overview of Payment System Architecture

Hassan et al. (2020) discovered that to meet the increasing demand for online transactions and purchases, improving electronic payment systems with new technologies has become a priority. Currently, online transactions are typically conducted through payment gateways. A payment gateway is a service provider that facilitates transactions between customers, sellers, and banks via the Internet. Secure transactions are made possible by the payment gateway, which encrypts sensitive data and makes sure that information is securely transported between customers and data processors. To mitigate risks associated with identity theft, email fraud, and to establish a secure system, Izhar et al. (2011) proposed a method using the Triple Data Encryption Standard (TDES), commonly known as 3DES, to encrypt transaction information and enhance transaction speeds within the gateway.

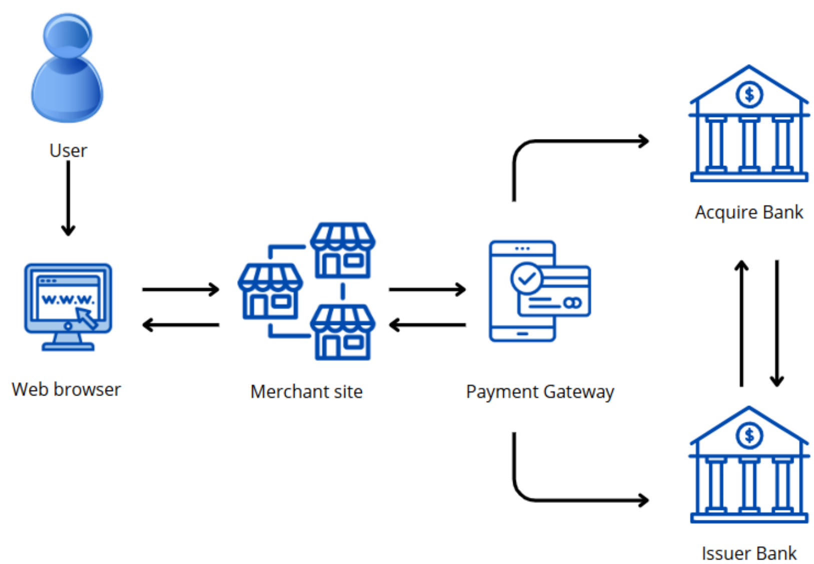
Additionally, there is another widely used encryption system in payment systems known as the RSA-Electronic Security System (ESS) (Pandey, 2018). This system was developed to address security and privacy issues concerning credit card information in e-commerce transactions. This model serves solely the purpose of ensuring security and privacy for processed transactions.

Musaev et al. (2018) researched privacy protection in online banking in Oman, and a secure electronic payment method was identified. The authors suggested using electronic tokens to facilitate digital payments, while also presenting an intermediary entity in this method to regulate transactions between payers and payees.

Another payment gateway proposed by Zay Oo (2019) allows direct transmission of transaction information (including credit or debit card details) to a transaction gateway, referred to as a Trusted Third Party (TTP), rather than directly through merchants on the Internet. This method is designed by incorporating secure socket layers (SSL) and RSA to enhance the supplementary relationship in the payment process. This form provides certain security benefits and minimizes information leakage and unauthorized access. Relevant client data is encrypted and safeguarded by using a secure payment channel, guaranteeing that only the legitimate payee can access the information. Another method based on the RSA algorithm is to generate a unique identifier for the mobile environment to prevent fraud on e-commerce platforms, allowing users to connect directly with merchants (Hajira Be et al., 2018).

In the research by Hassan et al. (2020), the authors proposed an architecture for the payment system on e-commerce platforms, involving participants such as Customers (C), Merchants (M), Payment Gateways (PG), User Banks (B), and Merchant Banks. This architecture was proposed with a focus on the safety and security of the system. The proposed system does not require customers to input personal identification information and generates a temporary identity to process service requests. The system process begins with each entity registering with the payment gateway to create a secret key. These secret keys are essential for securing communication information. Customers and merchants will create their secret keys, and after a customer requests to purchase a product, based on the temporary identity on the merchant's website, the merchant will send the request to the payment gateway. The payment gateway will perform several verification steps and forward the registered request to the customer. Simultaneously, the payment gateway will send encrypted information to the server. The customer's bank, upon receiving the request, will confirm the deduction of funds and send a response back to the payment gateway.

The payment gateway will perform calculations to determine the outcome and forward the results to the bank. The bank will receive the various transaction sessions and respond with the amount, and these responses will be accepted after verification.



Source:By Author

Figure 2.1. Architecture of the Payment System on E-commerce Platforms

According to the described system architecture, each participating entity will register with the payment gateway to generate secret keys, with customers and merchants creating secret keys between themselves. Subsequently, customers will connect their temporary identity on the merchant's website to place orders, and RSA encryption will be employed to encrypt the customer's card information after the order is placed. Once the order is successfully placed, the merchant will forward the encryption and decryption processes to the payment gateway. The customer's bank and the customer will then use RSA signatures to sign the document using the private key. In this process, the payment gateway is responsible for executing verification steps (encryption, decryption, and authentication) and sending decrement requests to the issuer.

2.2.2. Payment reconciliation system in affiliate marketing networks

When advertisers use affiliate marketing networks, the party responsible for tracking transaction data associated with the links is identified (Amarasekara & Mathrani, 2016). The tracking process is typically recorded based on the link's identifier, transaction time, transaction code, and transaction amount. In the AMNSTE (Affiliate Marketing Network simulation and testing environment), the affiliate network is designed using a service-oriented architecture (SOA) and has a modular design. The primary task of the platform is to track clicks and conversions, which is accomplished using a shared processing protocol (.ashx) without a user interface. However, depending on usage requirements, the system is designed to re-track clicks. Affiliate networks also provide a comprehensive API that allows advertisers to retrieve and edit tracking data. User-related information and referral data are stored in a Microsoft SQL database. Additionally, the system will store Campaign IDs and Revenue Models (CPC, CPA, revenue sharing, etc.). The reconciliation system is primarily focused on reconciling actions performed in affiliate marketing and extracting data for third-party tracking.

In the research conducted by Marten Weijer (2019), the author provided a detailed description of the Functional Architecture Model (FAM) for affiliate marketing networks. The functional architecture model is a comprehensive representation of a system's functions by presenting a hierarchical structure of modules. The affiliate marketing system includes modules such as Tracking Management, Affiliate Programs, Leads, Transactions, Performance Reporting, Payments, and Fraud Prevention. The transaction module contains a component that stores all transaction data collected from the leads module and a Publisher Matching component. This component matches a transaction with a lead and a producer. The part that ensures all transactions are authenticated and recorded securely, transparently, and reliably is managed by the Transaction Authentication module. The Payment module stores transactions on the blockchain and uses smart contracts to process affiliate payments. The module also includes a Payment Provider component and a Balancing component that manages payment information and the balances they wish to disburse for contracts. The smart contract module is linked to the blockchains of the transaction modules, and these contracts will be executed once a transaction occurs and is validated. The contracts are connected with payment service providers to ensure that the process is transparent.

The design of a model like that proposed by Marten Weijer (2019) can address several issues related to transparency, security, and privacy. However, this approach ignores the payment reconciliation concerning the commissions that the producer would pay to the publishers through this network, only guaranteeing transparency for transactions that take place within the affiliate marketing network.

2.3. Research Hypotheses

Based on the reviewed results, the author finds that the payment reconciliation system for affiliate marketing still has noteworthy points for research.

Hypothesis 1: Reconciliation activities typically occur regularly each month (Edelman & Brandi, 2015). The reconciliation and payment of commissions to publishers are usually conducted monthly with varying amounts.

Hypothesis 2: Traditional processes often carry risks and errors from manual reconciliation (Inampudi & Associates, 2023). Traditional commission payment reconciliation processes often consume significant time and resources each month. Current processes frequently encounter errors, overload, and delays when issues arise, leading to late payments.

Hypothesis 3: Providing commission-related information to publishers is necessary (Chattopadhyay, 2020). Publishers need to understand the affiliate programs they participate in and the commissions and compensations received to ensure their rights.

Hypothesis 4: The reconciliation system must ensure transparency and security (Marten Weijer, 2019). The recording of commissions between merchants and the distribution of commissions to publishers still lacks transparency and security. The lack of communication and exchanges between merchants and publishers leads to concerns regarding commission distribution within the network.

Hypothesis 5: Research focused on the reconciliation system between publishers and networks is relatively limited. Studies within the affiliate marketing network often concentrate on developing systems for reconciling marketing actions (Amarasekara & Mathrani, 2016) or reconciling transactions between merchants, publishers, and networks (Marten Weijer, 2019).

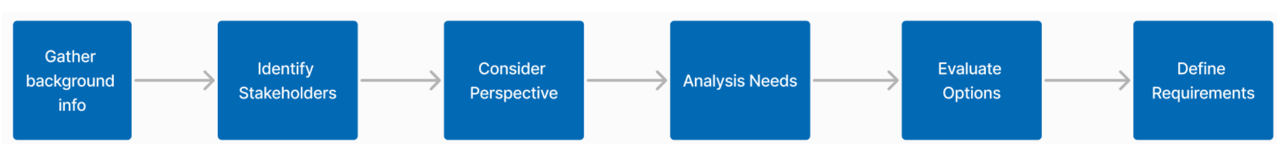
3. RESEARCH METHODOLOGY

3.1. Research process

The research focuses on the in-depth development of the reconciliation and payment module within the affiliate marketing system. This is achieved through an analysis of platform systems and interviews with experts industry who are involved in the affiliate marketing program on the TikTok Shop e-commerce platform. Interviews with experts are considered a way to quickly supplement foundational insights into the issues at hand. This survey aims to validate the information recorded from the literature review.

The survey was conducted over several meetings to clarify current processes, identify problems encountered in the application of traditional procedures, discuss considerations, and outline expectations for the upcoming system. From the interviews with experienced experts in affiliate marketing, the author has compiled the results into Appendix 1 to describe the topics discussed.

After confirming the issues related to the current reconciliation payment content, the author proceeded to investigate the stakeholders, analyzing in greater depth the usage needs and design options for a system that aligns with the actual business processes (Figure 3.1). The system design must ensure consistency with the current processes to avoid disrupting the operations of the network. Additionally, selecting the programming language, designing the data storage structure, and continuously monitoring the marketing activities of publishers are also crucial.



Source: By Author

Figure 3.1. Research process

Additionally, the author has compiled the current methods of calculating affiliate marketing commissions based on the commission amounts that merchants pay to the affiliate marketing network, the percentage commission rates, and the platform fees that publishers participating in the affiliate marketing network must pay. From this, a complete formula for the commissions of publishers within the network has been developed.

After conducting preliminary assessments, the author proceeded to analyze data from affiliate marketing activities to evaluate system usage traffic, the distribution proportions among payment methods, and refund rates. This analysis aims to propose suitable solutions to ensure privacy, stability, and security for all parties involved.

3.2. Research methodology

3.2.1. Use Case Diagram

This study will examine the Use Case Diagram model developed by Ivar Jacobson (1986), using collected information to describe the actors and objectives when building a system for the Pay per Sale (PPS) affiliate marketing model. This step is essential for clarifying issues and information in the early stages of the software engineering process and providing insights into how the affiliate marketing network operates. The UML (Unified Modeling Language) is a standard notation for modeling objects and systems in the real world while the Use Case Diagram is regarded as a subclass of behavioral diagrams. This diagram has characteristics that focus on describing the interactions between the system and external entities, organizations, or other systems. It is also considered a simple model, easy to understand, and accessible with straightforward symbols.

The Use Case Diagram features a simple structure with key symbols, including Actors - representing users or external systems, Use Cases - depicting the functions or actions that the system performs, and Relationships - the connections between Actors and Use Cases, such as association, include, and extend.

The use of this diagram helps stakeholders gain a clearer understanding of the system's functions and supports the identification and analysis of requirements, thereby providing useful information for the subsequent design phase.

3.2.2. Data Flow Diagram

After utilizing the Use Case Diagram model to identify stakeholders and their usage purposes, as well as to grasp the objectives when developing the system, the author will employ the Data Flow Diagram (DFD) to further investigate the interactions among stakeholders and the information flows within the system. The Data Flow Diagram helps delineate the boundaries of the system, providing a detailed representation of its components. This creates knowledge and offers an overarching view of the boundaries and connections.

In the DFD, specific symbols are defined for the diagram, such as rectangles, circles, arrows, and text labels that denote input data, output data, data storage points, and pathways between each destination. The DFD also provides information about the inputs and outputs of each entity without incorporating control flows, decision rules, or loops (Störrle, 2005).

In the process of calculating commissions for publishers, the author considers using data statistically gathered from the TikTok Affiliate Partner platform, which stores data on affiliate marketing activities on TikTok. The values that the TAP system can use to calculate commissions for publishers include the estimated order value (Est. commission base) and the Affiliate Partner Commission Rate (%). Subsequently, the network will also apply the commission rates payable for each product and the platform fees to compute the final commission for the publisher based on the following commission calculation formula after fees:

$$\boxed{\text{AFF CMS after fee} = \text{Est.commission base} * \text{Commission AFF } (\%) * (1 - \text{Network fee } (\%))}$$

Where:

- AFF CMS after fee: The commission after fees that the publisher receives.
- Est. commission base: The estimated order value.
- Commission AFF (%): The actual commission rate received by the publisher (the percentage of the commission does not include the minimum 1% fee).
- Network fee (%): The network fee when publishers participate in the network.

The commission, after being reconciled and marked as payment, will be displayed on the publisher's screen when they log into the system. The history of payment and reconciliation events will also be easily retrievable and reportable as needed. After successful payment, the system will record the transaction ID from the third party and automatically update the status of the payment. In the case of an order being canceled and requiring a refund, the system will automatically exclude that transaction from the commission calculation for the publisher.

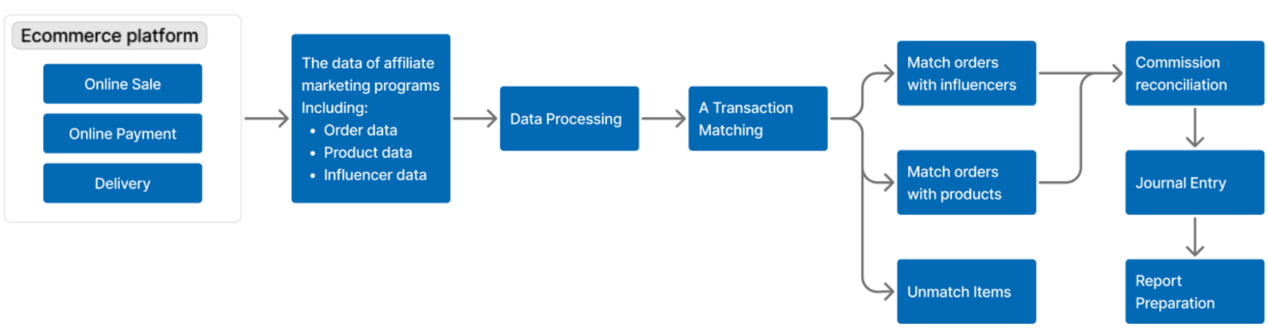
3.2.3. Business Process Modeling Notation (BPMN)

Business Process Modeling Notation (BPMN) is one of the modeling techniques used to describe business processes that was publicly released in 2004. BPMN describes with easily understandable symbols, serving business analysts and technical professionals who implement these processes (White, 2004). The BPMN model can include graphical elements such as Flow Objects, Connecting Objects, Swimlanes, and Artifacts. Additionally, BPMN may contain various adaptations aimed at simplifying and clarifying the description of business processes.

3.3. Research Data

The data from the TikTok Affiliate Partner platform provided aims to analyze and evaluate input information, thereby informing the system architecture that aligns with the company's current organization. Over five months, the company has supplied a rich dataset containing more than 5.900 orders from over 196 publishers in the affiliate marketing network, recorded from September 1, 2024, to December 28, 2024, all related to the affiliate marketing program on the TikTok Shop platform.

In addition, the company also provided data on product information, store details, and publicly available information such as usernames, IDs, and interaction counts of creators on the TikTok platform (Figure 3.2). This information is entirely public on the platform and does not contain any data protected by TikTok. These are actual orders that have been generated, thus they can be considered a representative sample to evaluate the operational processes and the effectiveness of the developed system.



Source: By Author

Figure 3.2. The process of data collection and processing

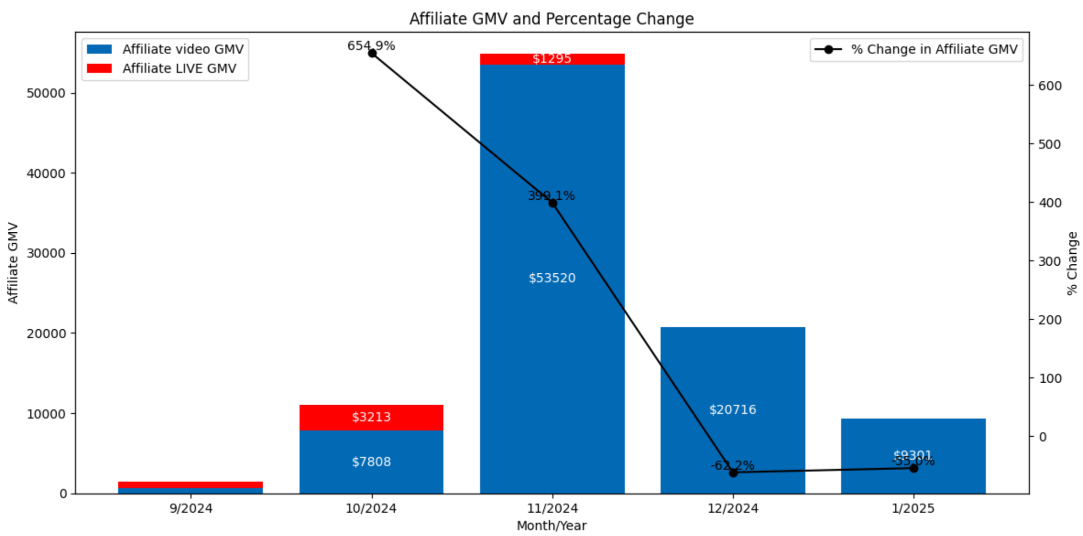
Additionally, the author has examined the timeliness and transparency of the data provided by the company. This includes comparing the collected information with the data publicly available on the TikTok platform to ensure that this information is accurate and reliable for analysis and decision-making.

Data regarding order values, product prices, and GMV affiliates in string format, collected via the API, will be cleaned and converted into numerical format. Any duplicate order IDs present in the extracted data will also be cleaned, retaining only one unique order ID. After processing, this data will be used to analyze order characteristics, transaction volumes, and payment methods to propose solutions that align with transaction flows and other characteristics. Furthermore, the author will encode the usernames and IDs of publishers to ensure security and privacy for the publishers in the network; therefore, the information of publishers in this study will not be identifiable on the TikTok platform.

4. RESEARCH RESULTS AND DISCUSSION

4.1. Data Description

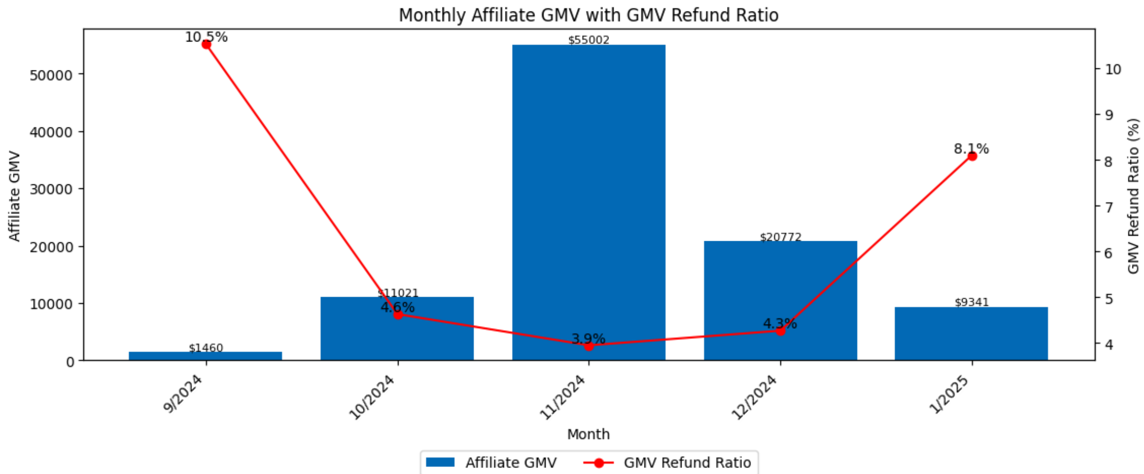
After analyzing the total estimated order value (Affiliate GMV) through affiliate marketing, the author noticed a significant disparity in revenue across the months. The month with the lowest revenue was September, with $1.459, while the highest revenue was recorded in November, totaling over $55.000, a 39-fold increase compared to October 2024. However, the proportion of affiliate marketing activity through live streaming sessions experienced a sharp decline from 29,15% (equivalent to $3.213) in October to just 2,32% (equivalent to $1.295) in November. In December and January, the total Affiliate GMV values were $20.772 and $9.340, respectively (Figure 4.1).



Source: By Author

Figure 4.1. Estimated order value in the affiliate marketing network from September 2024 to January 2025

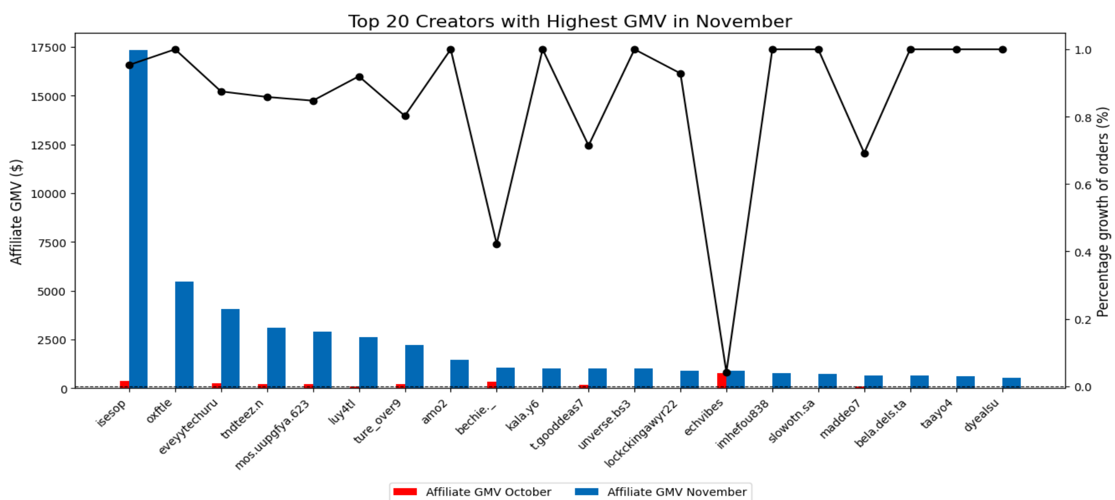
With fluctuating revenues and diverse affiliate marketing methods, there are significant concerns regarding the return rate of goods. In September 2024, the estimated order value was approximately $1.460, the lowest in the five-month study; however, the return rate for that month was the highest compared to the other months, at 10,5% (see Figure 4.2). Due to the reconciliation payment process occurring bi-monthly (approximately every 15 days), when the network implements the CPS or PPS model, it is necessary to consider the return rates of orders to establish an appropriate reconciliation system. This will help limit cases where commissions are calculated for returned orders, preventing losses for merchants and the marketing network.



Source: By Author

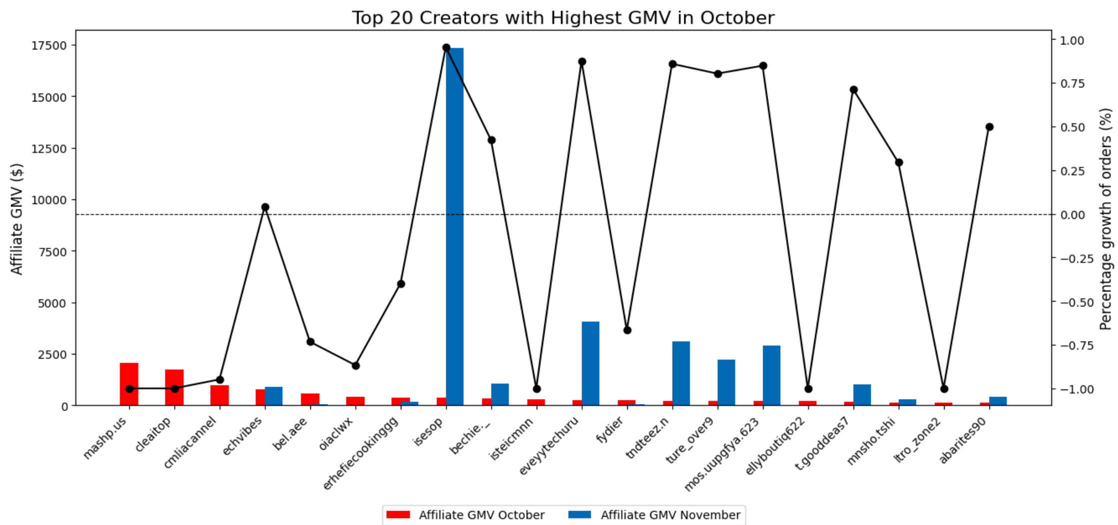
Figure 4.2. Return rate from September 2024 to January 2025.

When analyzing the total number of orders (including those paid and returned) from the top 20 creators participating in the affiliate marketing program in October (Figure 4.3) and November (Figure 4.4), a significant disparity in revenue among creators within the same network was evident. Some creators demonstrated performance that far exceeded that of others. Additionally, the revenue of the creators was not stable across the months; those who were in the leading group this month may not necessarily remain in the top group the following month, and some creators experienced remarkable growth percentages monthly.



Source: By Author

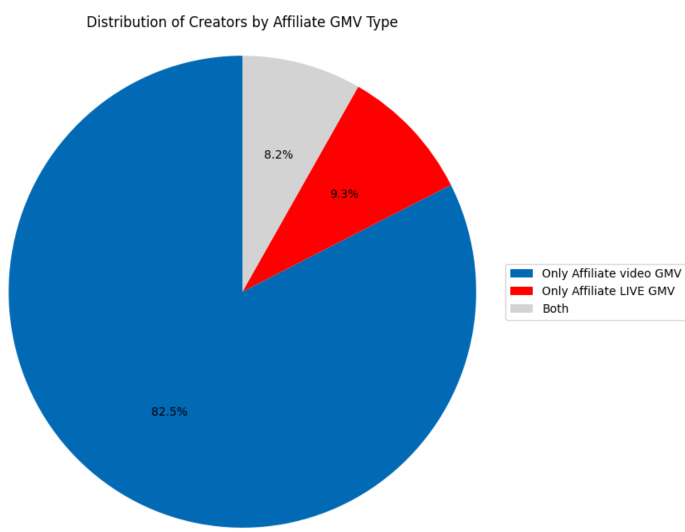
Figure 4.3. The 20 creators with the highest order values in October.



Source: By Author

Figure 4.4. The 20 creators with the highest order values in November.

Upon further investigation into the proportions that affiliate marketing programs typically apply, the author found that over 81,6% of creators focus solely on marketing by linking to products in their videos. The order rate generated by publishers who only market through live streaming is 9,3%, while the percentage utilizing both methods on the TikTok platform accounts for approximately 8,2% (see Figure 4.5).



Source: By Author

Figure 4.5. The proportion of affiliate marketing types in the network.

It can be observed that affiliate marketing transactions exhibit significant monthly fluctuations with diverse marketing types. Describing the data provides an overview of the characteristics of transactions on the TikTok Shop when purchasing through the affiliate marketing program introduced by publishers. This leads to the development of appropriate system design solutions to ensure traffic flow, stability, and feasibility for the reconciliation payment system of the affiliate marketing network.

4.2. Descriptive statistics

Table 4.1. Descriptive Statistics Table

Variables	**Obs**	**Mean**	**Std**	**Min**	**Max**
Affiliate GMV	5499	17,75	240,34	0	15736,13
Affiliate video GMV	5499	16,72	238,42	0	15736,13
Affiliate LIVE GMV	5499	0,98	28,97	0	1731,25
Orders	5499	0,58	8,03	0	537
Items sold	5499	0,60	8,18	0	545
GMV (refund)	5499	0,81	12,05	0	721,42

Source: By Author

Table 4.1 summarizes the descriptive statistics of the estimated order values with 5.499 observations in the network through affiliate marketing campaigns, including values such as mean, standard deviation, minimum, and maximum. The results indicate that the total estimated order value (Affiliate GMV) of one publisher in an affiliate marketing campaign reached a maximum of $15.736,13, derived from marketing through links in videos (Affiliate video GMV). The average total order value for the video link method is $16,72. Additionally, the total estimated order value through live streaming in an affiliate marketing campaign peaked at $1.731,25.

Moreover, the standard deviation of Affiliate video GMV and Affiliate GMV is notably high, exceeding 200, which indicates significant revenue fluctuations among publishers. The number of orders also varies from 0 to 537, suggesting that some publishers had no orders at all, while others had many. Furthermore, the highest total value of returned orders was $721,42. This high return rate may partly reflect the performance of publishers in marketing products on the e-commerce platform, such as misleading advertising or poor-quality products. This also provides a cautious perspective when designing the reconciliation and commission payment system within the network.

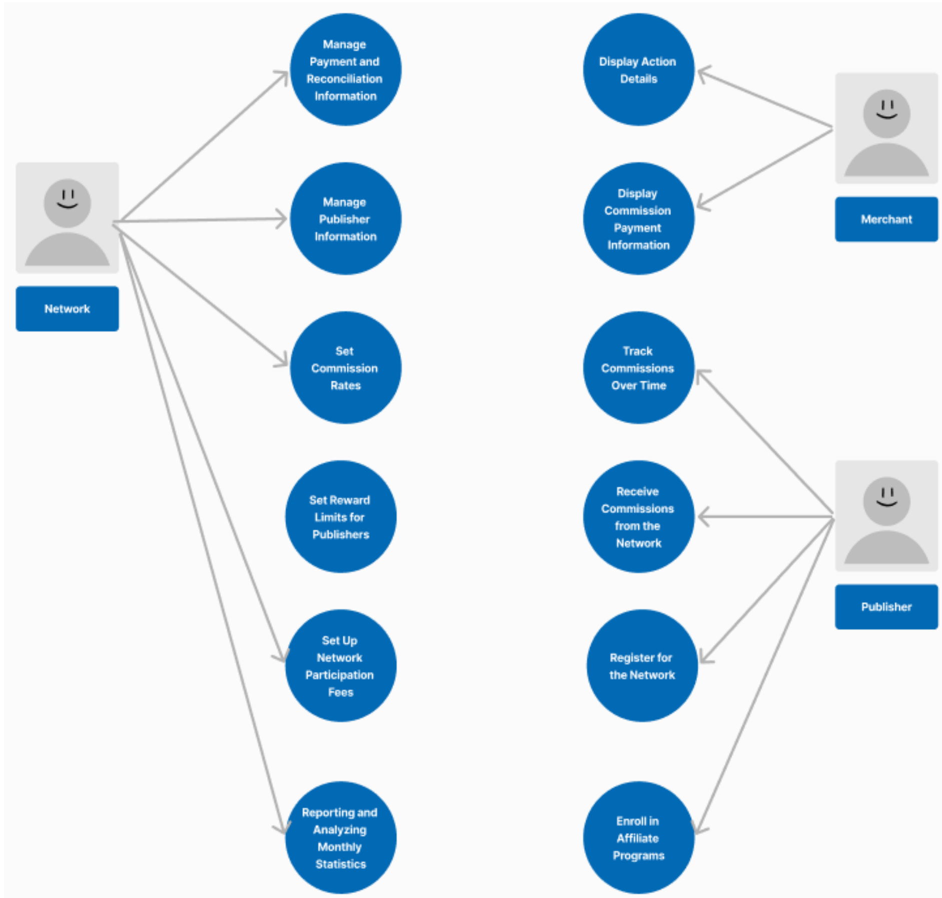
4.3. System Model Results

For affiliate marketing activities, the diagram will illustrate the interactions between the participants and the system (see Figure 4.6). The model will consist of three actors mentioned by the author Messer (2006). Each actor corresponds to a stakeholder involved in the payment reconciliation activities of the affiliate network.

Publisher: The publisher is viewed as an agent who uses a website to track commissions earned from promoting purchases made by customers on merchant sites. The main activities of the publisher in the network include: tracking commissions over time, receiving commissions from the network (Edelman & Brandi, 2015), registering with the network and enrolling in affiliate marketing programs (Papatla & Bhatnagar, 2002).

Merchant/Advertiser: The merchant is a participant in the affiliate marketing network seeking publishers to promote products and services on e-commerce platforms. Merchants typically review information about action details (Stokes, 2011) and the commissions to be paid.

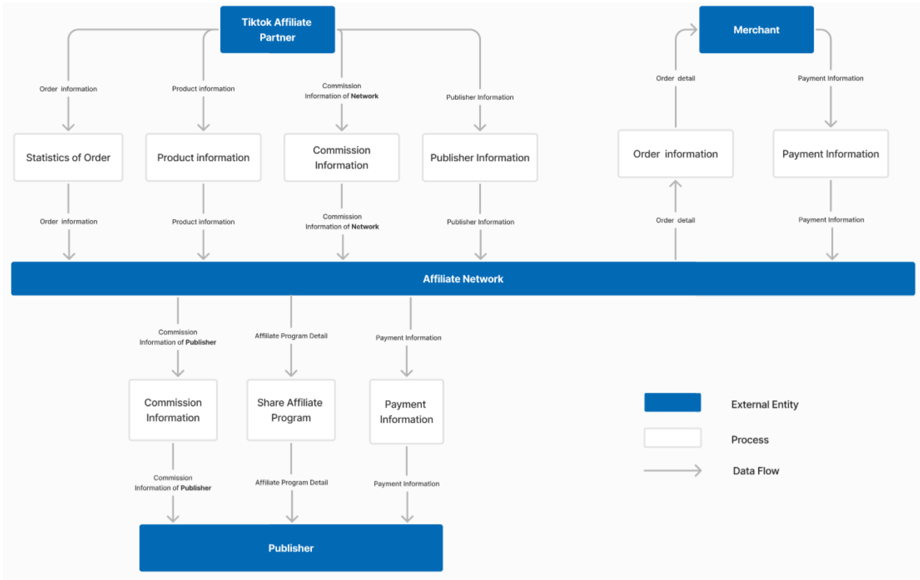
Network: The network acts as a third intermediary connecting merchants and publishers; it is also the entity that manages payment reconciliation in affiliate marketing campaigns for publishers. The main activities of the network within the system include managing payment reconciliation information for all publishers in the network, managing publisher information, establishing commission percentages based on each affiliate product, setting reward limits for publishers, and monthly reporting and statistics on commissions and publisher activities.



Source: By Author

Figure 4.6. A use case diagram for a payment reconciliation system

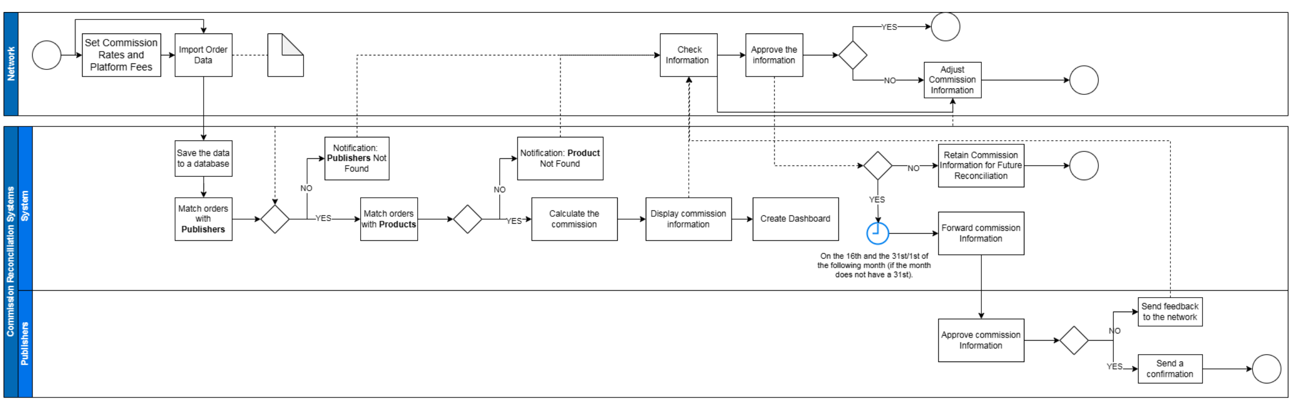
Additionally, the study presents a Data Flow Diagram (DFD) as described in Figure 4.7 primarily focused on analyzing the detailed data flows for the system. This analysis is based on components that include external entities such as individuals, departments, organizations, or data providers, along with processes, data flows, and data stores. A detailed description of the components and system processes is provided in Appendix 2. The processes in the DFD primarily focus on analyzing the detailed data flows for the reconciliation payment system when conducting affiliate marketing on the TikTok Shop platform.



Source: By Author

Figure 4.7. Data Flow Diagram for the reconciliation payment system.

Based on the Use Case Diagram and Data Flow Diagram, the author clarifies the process of using the system for reconciliation activities, as presented in Figure 4.8, through the Business Process Modeling Notation (BPMN) model. The BPMN model provides a deeper understanding of the process of reconciling commissions within the affiliate marketing system.



Source: By Author

Figure 4.8. The BPMN model describes the commission reconciliation process.

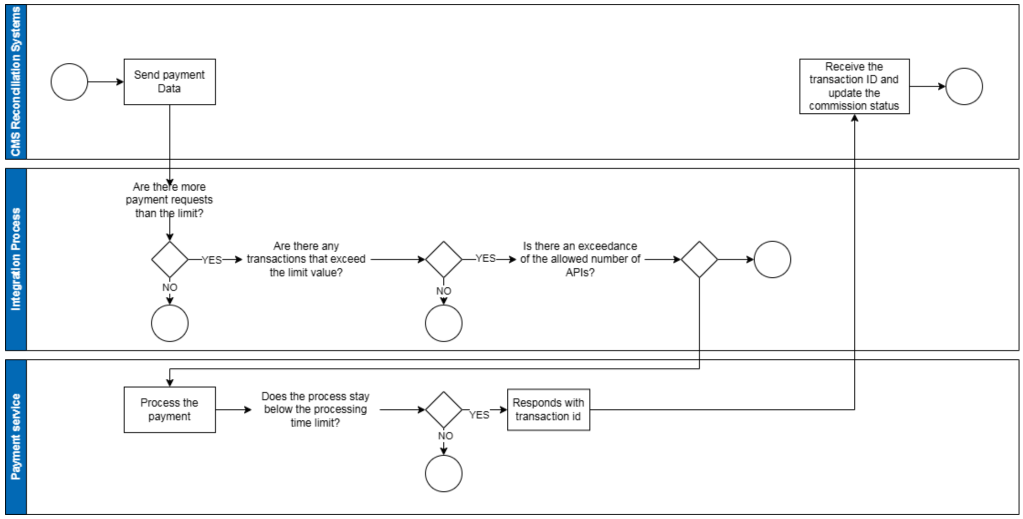
The initial reconciliation payment system requires the network to input the corresponding commission rates for each product, platform fees, and information about participating publishers by uploading files in formats such as XLSX, XLS, CSV, or via API

(Application Programming Interface) protocols. This data entry process may only need to be done once when using the system or when adjustments to the commission rates are necessary. Once the information is successfully entered, the network will upload the order data to the system or provide an API from TikTok Affiliate Partner (TAP) to directly record order information in real-time. TikTok Affiliate Partner (TAP) serves as a data center to capture centralized marketing information, with the data being recorded both comprehensively and in real time.

After capturing the generated order information, this data will be transmitted to the reconciliation payment system, which will automatically recalculate the commissions for the publishers. The order information will be matched against the information of publishers and products. If the order information matches the existing product and publisher details, the system will proceed to calculate the commission for that order and generate a report. In cases where the information does not match, the system will notify and prompt the network to verify the information.

The results of the commission calculations will be displayed on the administrator's homepage, allowing the affiliate network administrator to check the system's update status. In the reconciliation time for the publisher, commissions that have been confirmed and approved by the network without any issues will be shown. At the end of the reconciliation period, on the 23rd or 8th of each month, the system will notify publishers whose commissions meet the minimum requirements set by the network to initiate the payment process. If a publisher does not meet the minimum commission threshold, the system will retain this commission and carry it over to the next reconciliation period.

A list of publishers eligible for commission payments will be displayed, allowing the network to process payments based on this list. The system will record the commission payment status for the publisher based on the transaction code sent back through the API connection of third-party payment service providers, such as Fintech companies or banks (see Figure 4.9). Then, the network can also track the payment statuses, including "In Reconciliation," "Pending Payment," and "Paid".



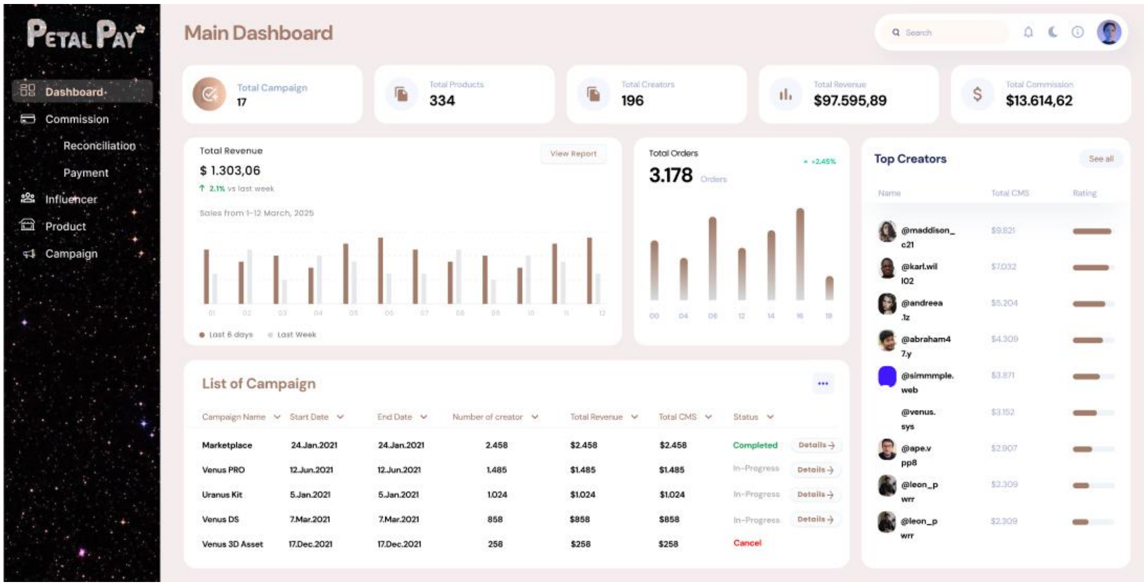
Source: By Author

Figure 4.9. The BPMN model describes the commission payment process

Additionally, each publisher can track the status of their reconciliation payments for each period by accessing the system with an account provided by the network. Commission information will be updated on the publisher's payment information page as soon as the network changes the payment status from "In Reconciliation" to "Pending Payment" or to another status. Similarly, merchants can also access the system to view the commission amounts they need to pay to the network after each reconciliation period. However, this requires the approval of the system administrator.

After each reconciliation and payment period, the network can monitor statistically aggregated data over time regarding the commissions received by publishers, the actual commissions received by the network, and the effectiveness of campaigns based on commission fluctuations in each reconciliation period. From the statistical data, managers can formulate appropriate strategies to improve business performance.

Based on the results of the BPMN system description, the author developed a website mockup that includes the essential features. The system includes modules such as Dashboard, Commission (with two tabs for Reconciliation and Payment), Publisher, Product, and Campaign, as shown in Figure 4.8.



Source: By Author

Figure 4.10. Commission reconciliation website interface in the affiliate marketing system

The reconciliation module is used to record the orders generated by publishers in the system, which are uploaded from files or transmitted via the TikTok API. From there, commissions are calculated based on the commission percentages of each product set in the Product module and the platform fees. Once the information has been verified, the network can proceed with making payments by sending batch payment requests, and changing the status of reconciliations from "In Reconciliation" to "Pending Payment." Simultaneously, these commissions will also be displayed in the Payment tab.

Upon successful payment, the system will receive feedback regarding the transaction ID to match and change the status of the paid commissions from "Pending Payment" to "Paid," as described in Figure 4.9. Any erroneous transactions will be flagged in the system for the network to review and retry.

The system will also automatically generate charts and reports based on the input data, displayed comprehensively on the Dashboard (see Figure 4.10). The Dashboard provides network managers with an overview of the performance of marketing campaigns, publishers, revenue, and commissions in real time. This enables the network to assess and make quicker, more effective decisions.

5. CONCLUSION AND RECOMMENDATION

5.1. Conclusion

This study provides insights into the operations of the affiliate marketing system on the TikTok Shop platform, particularly in managing commissions. Through data analysis from orders generated by affiliate marketing activities, significant discrepancies in revenue and performance were observed between months, as well as fluctuations in marketing methods. The system must also accommodate sudden increases in transactions while minimizing overload situations. Additionally, the noticeable revenue fluctuations highlight the importance of optimizing marketing strategies and selecting appropriate marketing forms to enhance revenue.

Moreover, when examining the return rates of activities, the author noted significant discrepancies between two months with nearly identical order values, with higher return rates occurring in months with lower order values. Based on the network's reconciliation characteristics, such as a 15-day reconciliation period each month and only applying to completed orders, the system must quickly and transparently update data to minimize errors in the commissions owed to publishers. Furthermore, improving the payment process and commission management is essential to reduce delays in commission payments to publishers.

In this study, the author also proposes developing a flexible, transparent, and more automated reconciliation payment system based on analyses from the Use Case Diagram, Data Flow Diagram, and BPMN. This system aims to meet reconciliation activities and enhance the operational efficiency of the affiliate marketing network. It can help networks automatically calculate commissions with continuously updated data via API protocols or by uploading data files, thereby minimizing resource use and ensuring transparency in the reconciliation and payment of publishers' commissions. This system not only supports creators in tracking commissions while participating in affiliate marketing activities on the TikTok Shop platform but also helps merchants and networks manage information more effectively.

Automatically generated and visualized data will also provide network managers with a clearer and more accurate overview of the performance of publishers in the affiliate marketing network. This allows for the establishment of bonuses or incentives for publishers based on the value they bring, thereby boosting publishers' morale when participating in the network. Additionally, the reports provided can assist managers in evaluating and making quick, timely decisions that enhance operational efficiency.

Finally, this study opens up avenues for future research aimed at improving marketing and payment methods in the e-commerce sector, to meet the growing demands of consumers and content creators on platforms like TikTok.

5.2. Recommendations

To optimize the reconciliation-payment system for affiliate marketing operations, the system needs to automate processes such as uploading input data files via API (Application Programming Interface) connections directly from the TikTok system. This method can significantly minimize the operational time within the system and increase the accuracy of the input data, thereby reducing errors in the commission information paid to publishers.

Furthermore, the system should enhance transaction security by using identification codes with RSA algorithms for commission payment transactions, encrypting data, and implementing two-factor authentication (2FA) to protect user accounts. The system also needs to develop flexible reconciliation forms with various payment methods, such as cash on delivery (COD) and buy now, pay later, to ensure that commissions are paid fully and accurately. Additionally, the stability of the system must be carefully considered, ensuring compatibility across different browsers such as Chrome, Firefox, and Safari, achieving good performance through tools like Google PageSpeed Insights or GTmetrix to measure page load speeds.

During development, user experience testing should be conducted using user behavior analytics (such as Google Analytics) to monitor traffic and interactions. A/B testing could also be implemented to compare the effectiveness of different versions of the website.

Future research may explore the application of artificial intelligence (AI) to evaluate and propose suitable operational scenarios and strategies based on historical commission data for managers or publishers. Publishers could also utilize AI assessments to determine appropriate development directions for their affiliate marketing activities.

5.3. Limitations of the Study

Although this study has provided an overview of the operations of the affiliate marketing system on the TikTok Shop platform, several notable limitations still exist.

Firstly, the research did not delve deeply into the data processing system, which may impact the ability to aggregate and analyze information effectively. A lack of thorough investigation into existing data processing technologies and methods could result in shortcomings in assessing the accuracy and reliability of the collected data.

Secondly, the study did not consider the technology acceptance capabilities of the stakeholders. This could lead to obstacles when developing and implementing the system for management units, merchants, and publishers. Assessing technology acceptance could be applied using the Technology Acceptance Model (TAM), which helps predict and explain approximately 40% of the variations in usage intention and actual use.

Thirdly, the system lacks high security. Insufficient security could lead to issues related to information theft, fraud, and other risks. This is a significant issue that needs to be addressed when building a reconciliation-payment system for a network with multiple stakeholders.

Finally, there will be a certain degree of delay in data updates within the system. Data transmission between parties through the API protocol will still take some time and may encounter errors during the data transfer process.

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APPENDIX

Appendix 1. Results of Recorded Opinions

Hypotheses	Results	Expert Insights
H_1	Supporrted	Currently, the commission management process for partners is conducted twice a month (with each reconciliation period lasting 15 days). Specifically: • The first period runs from the 8th to the 23rd of each month. • The second period runs from the 23rd of this month to the 8th of the following month. The affiliate marketing network processes payments for partners twice a month, on the 16th and the 31st (if the month has 31 days) or on the 1st of the next month (if the month does not have 31 days).
H_2	Supporrted	The payment and reconciliation process is often manually tracked using tools like Excel, which consumes a considerable amount of time and resources to carry out reconciliations for the network.
H_3	Supporrted	The commission distribution process for partners is primarily managed and distributed by the affiliate marketing network. On the TikTok Shop system, only 1% of the commission is recorded as a formal representation, while the actual commission for partners is calculated based on the commission received by the TikTok Partner Affiliate (TAP) network.

		When an order is canceled, the individual responsible for payment reconciliation must manually verify the details. In cases where an order is canceled or refunded close to the reconciliation/payment date, the commission for that order will be deducted in the following month. Therefore, accurate commission reconciliation is essential to ensure the rights and benefits of publishers within the network.
H_4	Supporrted	The system being developed must ensure accuracy and data security for the network.
H_5	Opposed	Currently, there are not many studies related to the reconciliation payment system for affiliate marketing activities; however, there is enough information to support the development of the system.

Source: By Author

Appendix 2: Description of Processes in the Data Flow Diagram (DFD)

Process	**Description**
Share Affiliate Program	The network sends a list of products to publishers for promotion (Walker et al., 2010)
Sale Information	Information about orders placed through the publishers' links.
Commision Information	The commission paid is based on the revenue from the products sold.

Product Information	Details of products that have been tagged for promotion through affiliate marketing links.
Publisher Information	Information including Username, First Name, and Last Name used for account registration on the TikTok platform.
Statistic of sales	Order information including Order ID, Product Name, SKU, Product ID, Price, Quantity, Store Name, Store Code, Order Status, Content Type, Estimated Order Value, Estimated Affiliate Commission Rate, Creator Commission Rate, Actual Order Value, Return Quantity, Refund Quantity, Order Creation Time, Delivery Time, Payment Time, Payment ID, and Payment Method.
Payment Information	Payment information disbursed for the affiliate marketing program on a periodic basis.

Source: By Author